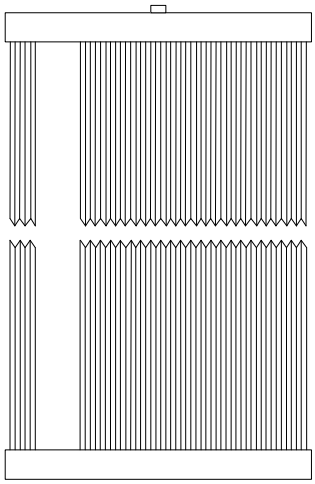
SIEMENS

Installation Instructions
Model PSC-ISO-CBL

PSC-12 Isolating Cable

Introduction

Depending on the system configuration, the 100AH maximum battery size for a single PSC-12 may be insufficient to support the desired equipment in a given enclosure. Using the model PSC-ISO-CBL, a second PSC-12 can be installed, eliminating the need to add a second CAB-1/-2/-3 enclosure and the NIC-Cs needed to interconnect them. The PSC-ISO-CBL cable isolates the 6.2V and 24V power supplies between the PSC-12s, which are sharing the same backplane. Depending on the size of the batteries for each PSC-12, either one or two battery enclosures will be required.



Zeus

Zeus 10 introduces the concept of a Primary PSC-12 and a Non-Primary PSC-12. Primary PSC-12s must be interconnected via the NIC-C. Non-Primary PSC-12s must be connected to a Primary PSC-12 via the PSC-ISO-CBL cable.

Note: A single enclosure with a Primary PSC-12 and a Non-Primary PSC-12 does not require a NIC-C.

In Zeus, each PSC-12, either Primary or Non-Primary, must be installed in its own enclosure. This is necessary for the proper calculation of power supply loading and battery size. In the actual installation, the Primary and Non-Primary PSC-12s will be installed in the same enclosure.

Note: Careful attention must be paid to the amount of equipment in the two enclosures. Zeus will not be able to detect that the equipment may not fit into the actual single enclosure.

There is no restriction as to which modules can be connected to either a Primary of Non-Primary PSC-12, however, in Zeus each primary PSC-12 (in a multi enclosure system, see Figure 5) must have a NIC-C in its enclosure.

Installation

Locate the Primary and Non-Primary PSC-12s adjacent to one another in the center of the CAB-1 or in the lower right hand corner of the CAB-2/-3. Install the PSC-ISO-CBL between theses two power supplies.

Note: It is not possible to split the power within a CC-5 of CC-2. All of the equipment within a single card cage must be powered from the same PSC-12.

Organize the equipment in a manner that makes it easy to identify which PSC-12 is providing power to which modules. A vertical split is recommended in the CAB-2, a horizontal split (by row) is best in a CAB-3.

Refer to Figures 2-5 for typical cable routing.

Ground Fault detection must be provided by only one PSC-12 in an enclosure. Disable ground fault detection on all Non-Primary PSC-12s by moving the jumper on P10 from the Enable position (1&2) to the Disable position (2&3). See Figure 1 for the location of P10 on the PSC-12.

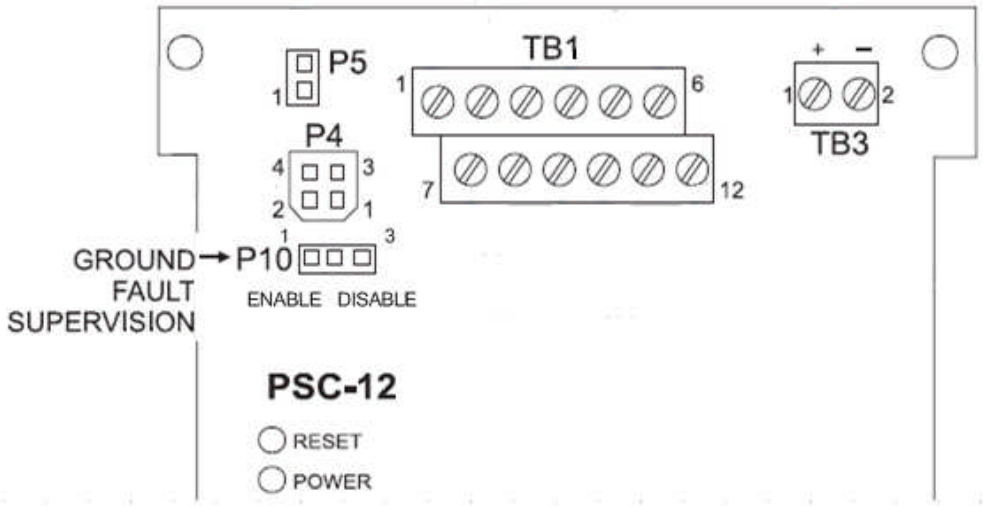
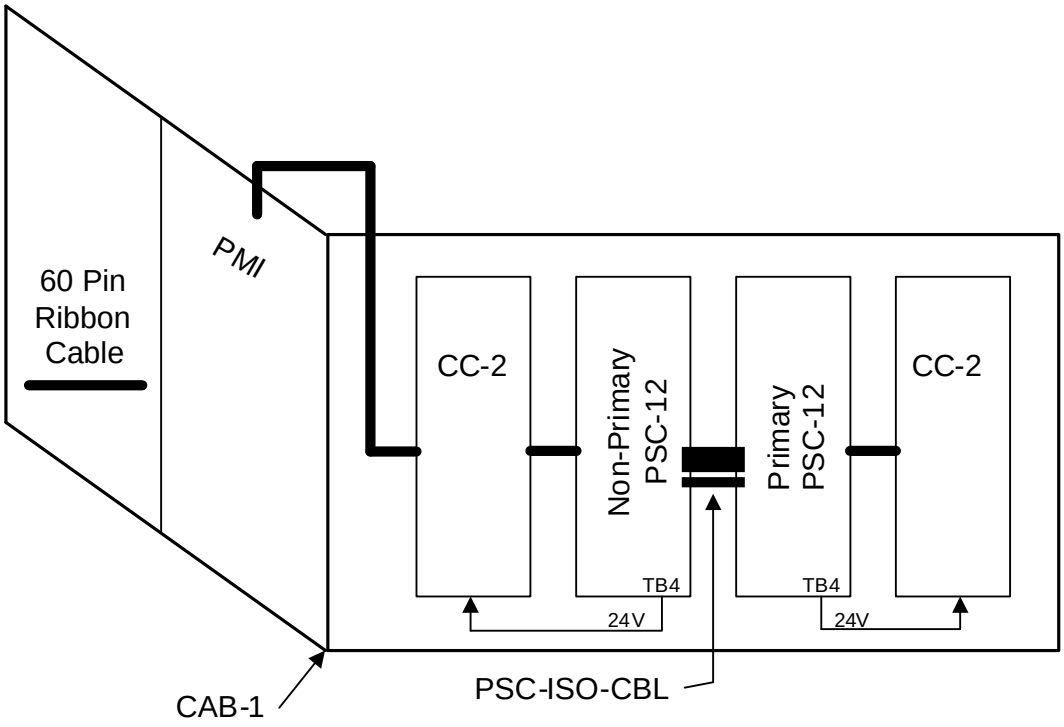
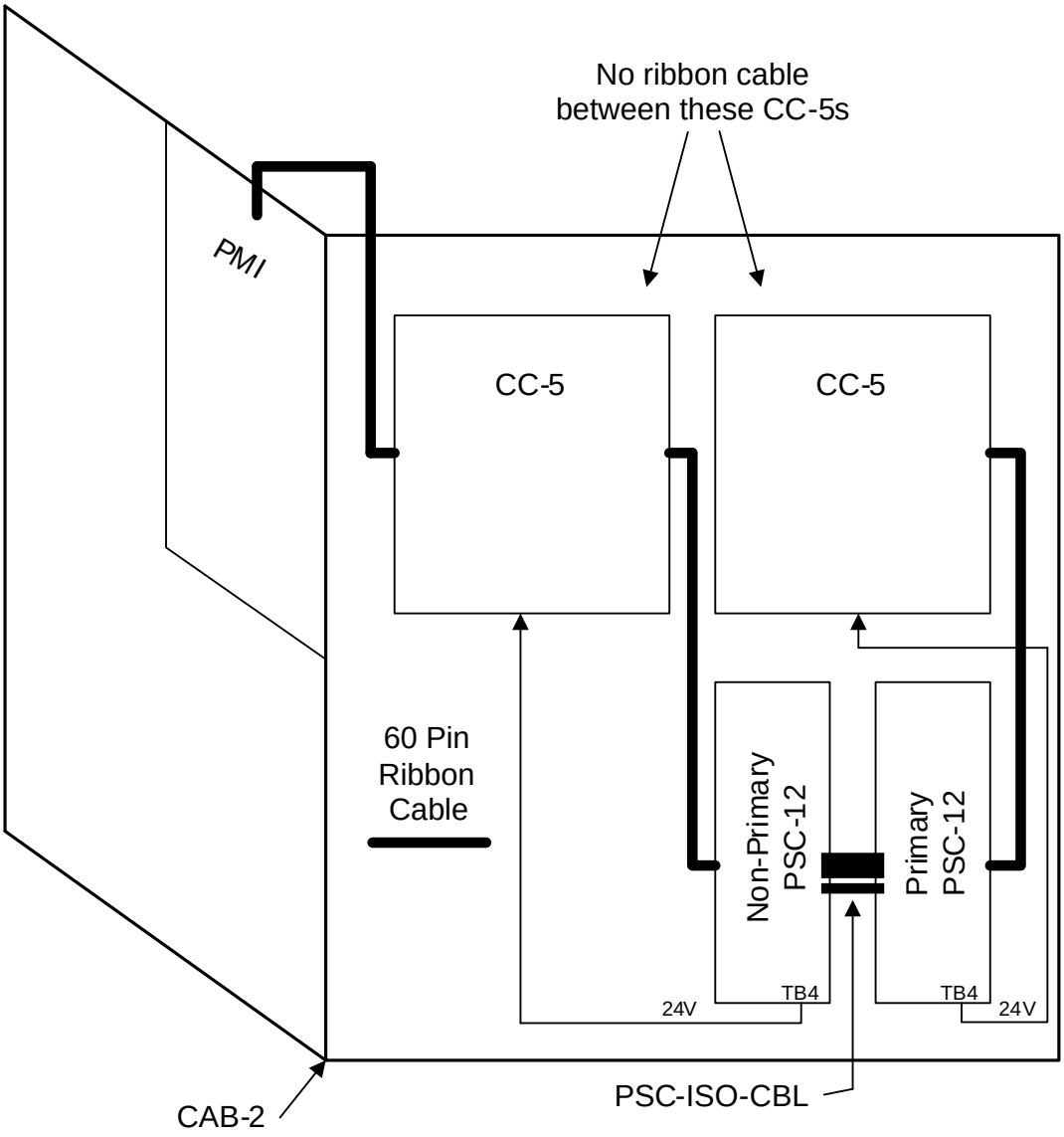


Figure 1: PSC-12 Ground Fault Enable/Disable Jumper P10 Location



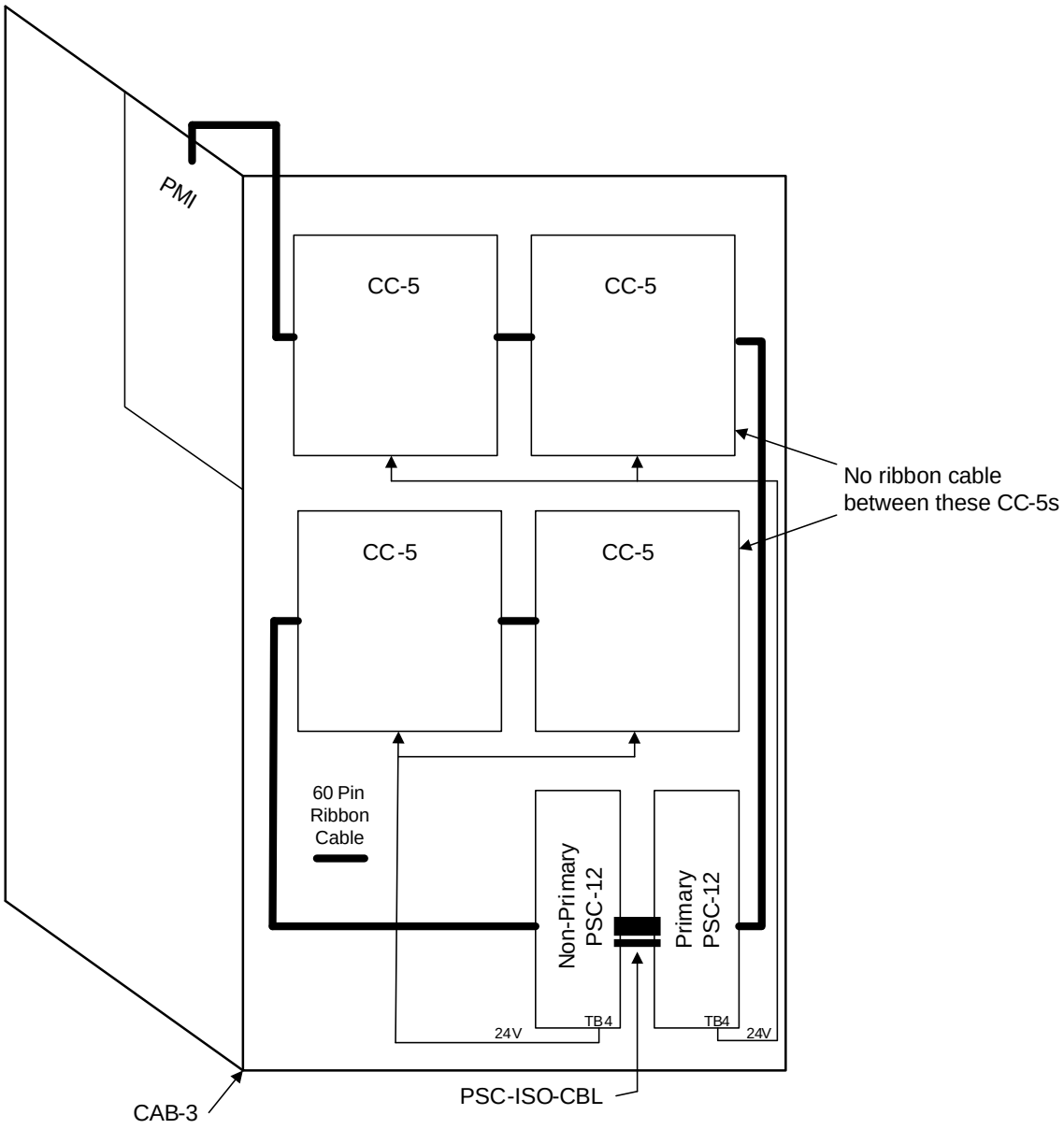
Note: Each PSC-12 requires its own battery set. Depending on the battery size either one or two battery boxes are needed.

Figure 2: Single CAB-1 enclosure



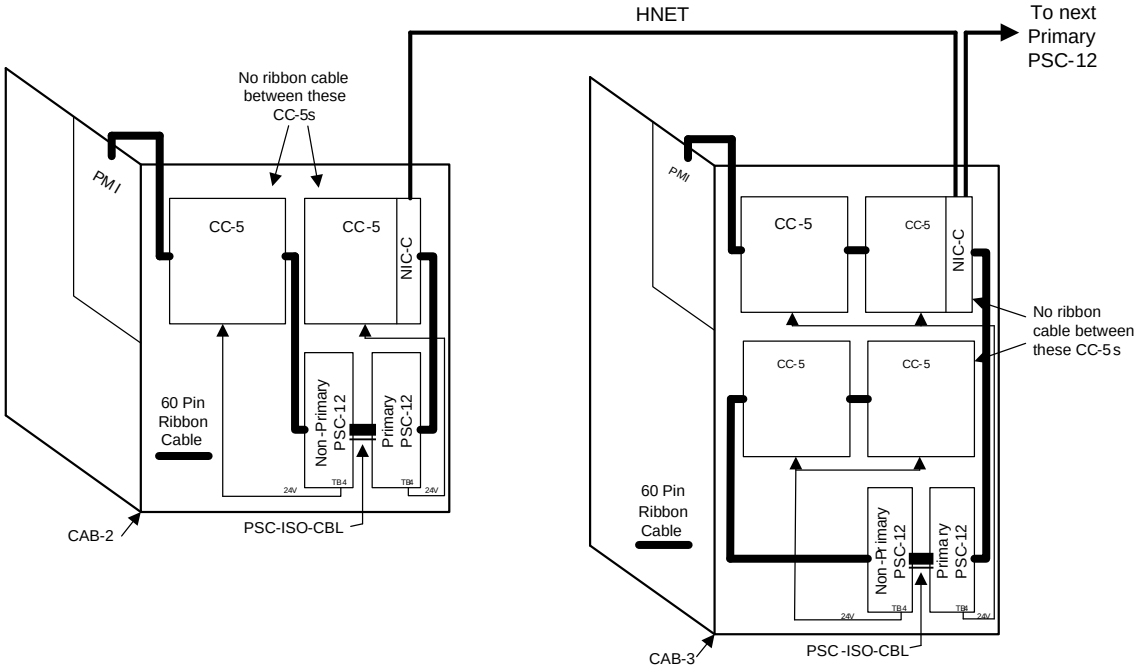
Note: Each PSC-12 requires its own battery set. Depending on the battery size either one or two battery boxes are needed.

Figure 3: Single CAB-2 enclosure



Note: Each PSC-12 requires its own battery set. Depending on the battery size either one or two battery boxes are needed.

Figure 4: Single CAB-3 enclosure



Note: Each PSC-12 requires its own battery set. Depending on the battery size either one or two battery boxes are needed per CAB enclosure.

Figure 5: Multiple enclosures

Separation of Power Limited and Non-Power Limited Wiring

Wire in accordance with local codes and Article 760 of the NEC (NFPA 70) latest edition. In compliance with NEC Article 760, all power limited fire protective signaling conductors must be separated by a minimum of ¼ inch from all of the following located within the control panel.

- Electric Light Circuits
- Power Circuits
- Class 1 or Non-Power Limited fire protective signaling conductors

Power Limited and Non-Power Limited circuits leaving the enclosure must be in separate conduits.

Refer to Figures 6-8 for the proper routing of Power Limited (Dashed Lines) and Non-Power Limited (Solid Lines) wiring within the CAB-1/-2/-3 enclosures.

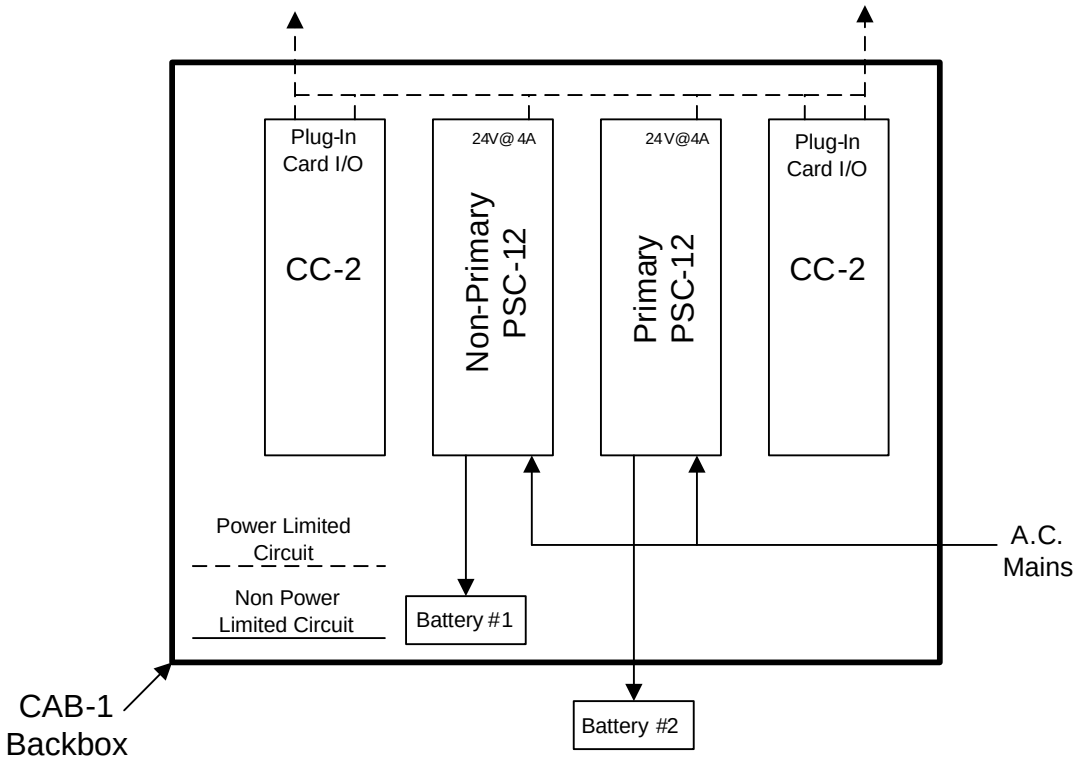


Figure 6: CAB-1

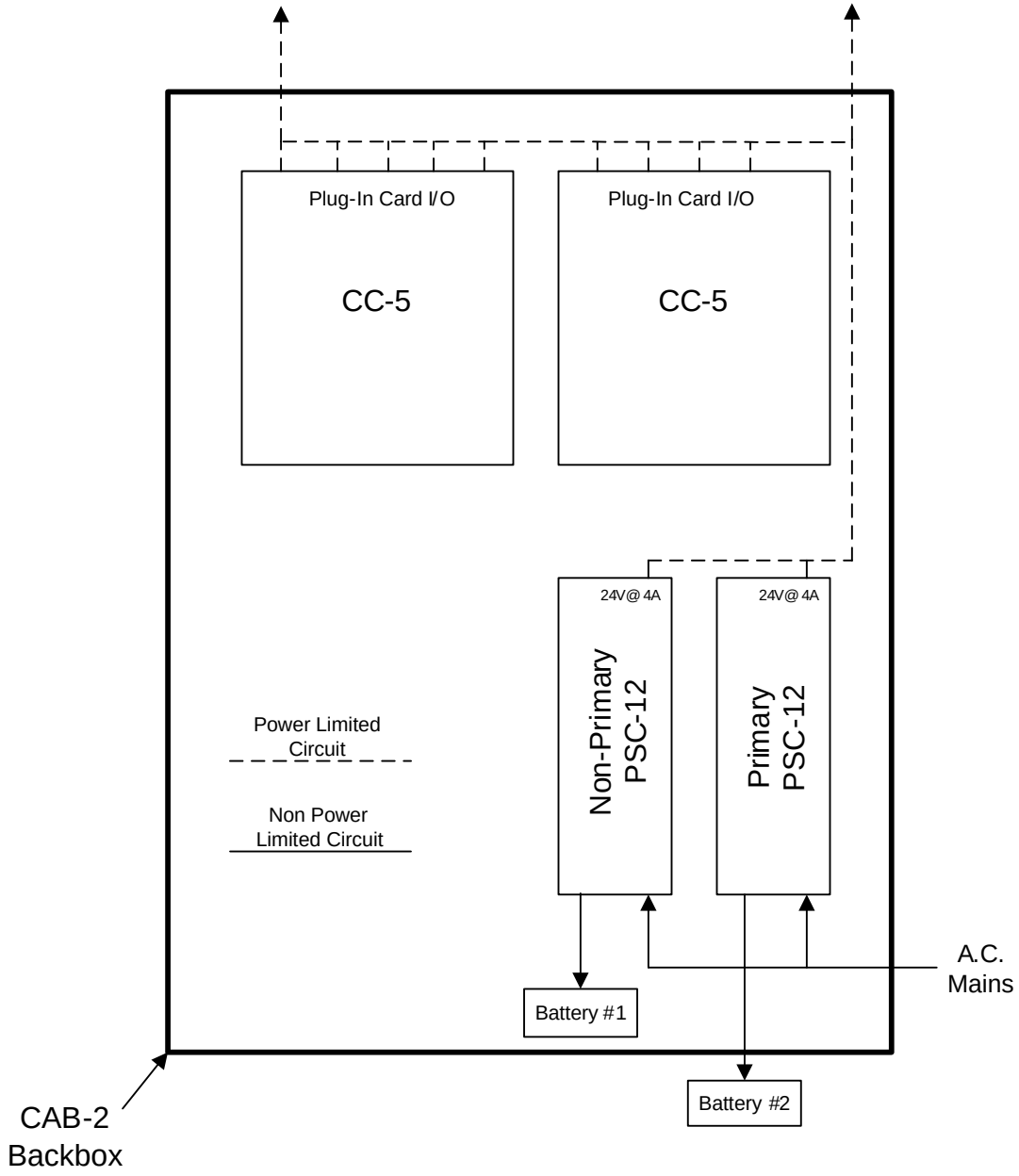


Figure 7: CAB-2

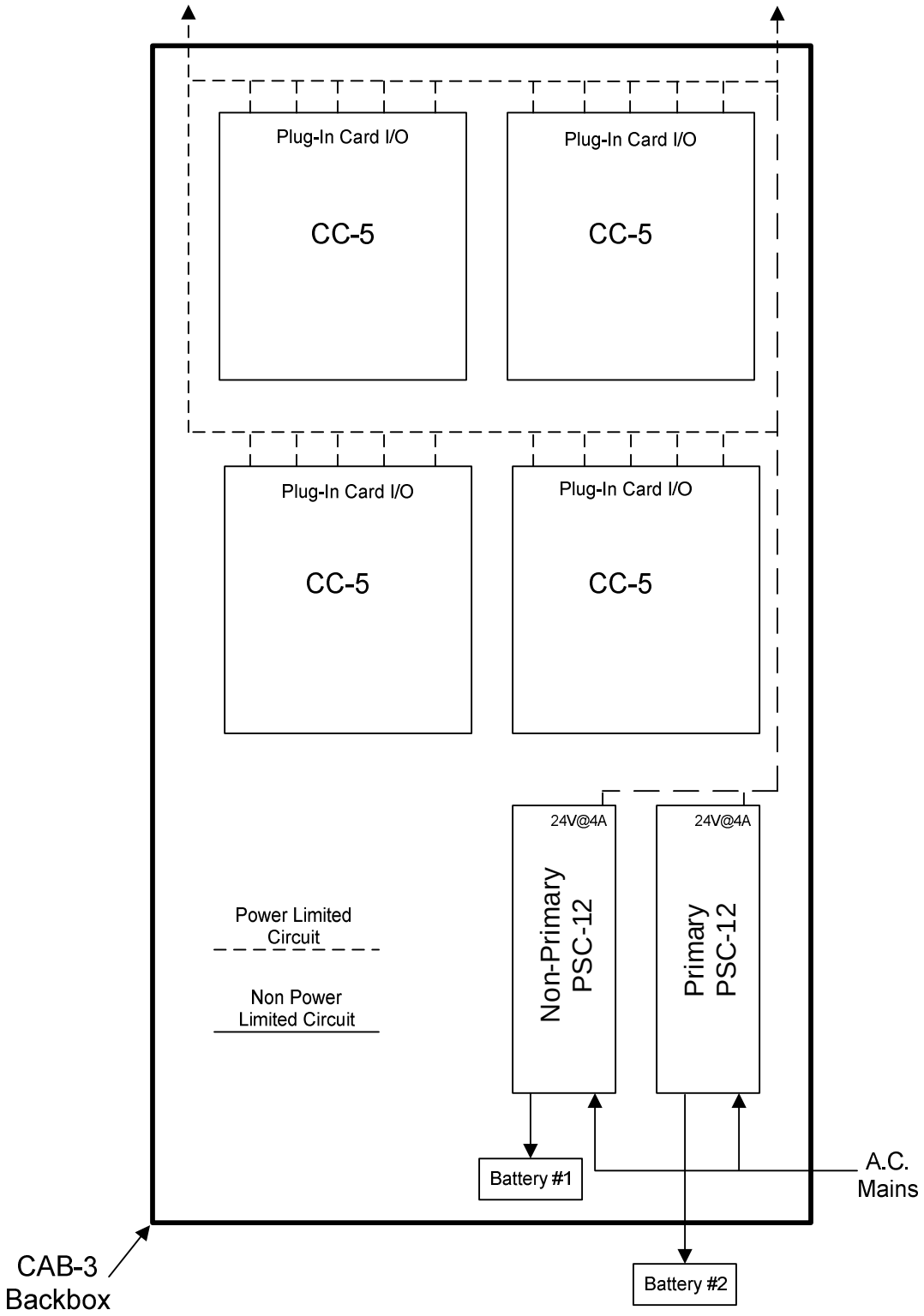


Figure 8: CAB-3

Electrical Ratings

ATTENTION: The total Standby current for the Primary and Non-Primary PSC-12s in a single enclosure must not exceed 5A and the total Alarm current must not exceed 12A.

Each PSC-12, either Primary or Non-Primary, will be located in a unique enclosure in Zeus. Zeus will perform battery size and power supply loading calculations independently. Perform the following steps to determine the proper battery size and alarm/standby loading for the combination of the Primary and Non-Primary PSC-12s that will be installed in a single enclosure.

1) In Zeus, generate a "System Power & Battery Calculation Report". This report is available under Tools -> Report Generation -> System Power & Battery Calculation. See the example below. Enclosure #1 contains a Primary PSC-12 and enclosure #2 contains a Non-Primary PSC-12.
2) Observe the calculated battery size for each enclosure. This is the required battery size for either the Primary PSC-12 (13.841AH in the example) or the Non-Primary PSC-12 (11.292AH in the example). Depending on the battery size either one or two battery boxes may be required.
3) For each pair of Primary and Non-Primary PSC-12s that will be in the same enclosure in the final system, add the calculated Standby currents together. In the example, the total is 0.296AMP + 0.101AMP = 0.397AMP. **The sum must not exceed 5A.**
4) For each pair of Primary and Non-Primary PSC-12s that will be in the same enclosure in the final system, add the calculated Alarm currents together. In the example the total is 4.453AMP + 6.258AMP = 10.711AMP. **The sum must not exceed 12A.**

Siemens Fire Safety - Project PSC-ISO-CBL
Power Load and Battery Computation Summary Report Printed on 12 Nov 2013 at 09:49:20

Supervisory Current and Battery Computation Summary
All Battery Selections PASS

Power Load Computation Summary
All Power Load Computations PASS

Supervisory Battery and Power Load Computation Summary

Enclosure	Battery Size	Back Plane	Standby	Alarm	6.2VDC
CAB-2 Enclosure #1	13.841AH	0.352AMP	0.296AMP	4.453AMP	0.200AMP
CAB-2 Enclosure #2	11.292AH	0.157AMP	0.101AMP	6.258AMP	0.200AMP

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